NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

**RACE** Regular Analysis through Continuous Exercise

**JEE (Main + Advanced) 2023**
**ENTHUSIAST + LEADER COURSE**

**SCORE ADVANCED** | **MATHEMATICS** | **M.M. 61**

REVISION ASSINGMENT # 03 | (COMPLEX NUMBER) | TIME : 60 MIN.

## SECTION–I

**Straight Objective Type** (3 Marks each, –1 for wrong answer)

**1.** All roots of the equation, $(1 + z)^6 + z^6 = 0$ -
(A) lie on a unit circle with centre at the origin
(B) lie on a unit circle with centre at (–1, 0)
(C) lie on the vertices of a regular polygon with centre at the origin.
(D) are collinear

**2.** If $P = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$, $a = P + P^2 + P^4$ and $b = P^3 + P^5 + P^6$, then the value of $|(a + b) + abi|$ (where $i = \sqrt{-1}$) is -
(A) $\sqrt{3}$ (B) $\sqrt{5}$ (C) $\sqrt{6}$ (D) 2

**3.** If $z_1, z_2, z_3$ are the roots of the equation $z^3 - z^2(1 + 3i) + z(3i - 2) + 2 = 0$, then $\text{Im}(z_1) + \text{Im}(z_2) + \text{Im}(z_3)$ is -
(A) 0 (B) 1 (C) 2 (D) 3

**4.** Number of values of z satisfying $|z - 2i| = 2$ and $z(1 - i) - \bar{z}(1 + i) = 4i$ simultaneously is -
(A) 0 (B) 2 (C) 3 (D) 4

**5.** Value of $\sum_{k=1}^{100}\left(i^{k!} + \omega^{k!}\right)$, where $i = \sqrt{-1}$ and $\omega$ is complex cube root of unity, is -
(A) $190 + \omega$ (B) $192 + \omega^2$ (C) $190 + i$ (D) $192 + i$

**6.** If $z_2$ & $z_4$ are conjugates of $z_1$ & $z_3$ respectively, then $\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right)$ is equal to -
(A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{2}$ (C) $\pi$ (D) 0

**7.** If $|z - 3 - 2i| = |z + 2i|$, where z is a complex number, then minimum value of $|z|$ will be -
(A) $\frac{1}{2}$ (B) $\frac{4}{5}$ (C) $\frac{7}{10}$ (D) $\frac{9}{10}$

**8.** If one of the vertex of the regular hexagon, circumscribing the circle $|z - 1 - i| = \sqrt{3}$ is $2 + (1 + \sqrt{3})i$, then the complex number which is not representing any vertex of the hexagon is -
(A) $3 + i$ (B) $(1 + \sqrt{3})i$ (C) $-3 + i$ (D) $2 + (1 - \sqrt{3})i$

**Multiple Correct Answer Type** (4 Marks each, –1 for wrong answer)

**9.** If $|z_1| = 1$, $|z_2 - 2| = 3$ and $|z_3 - 5| = 6$, then maximum value of $|2z_1 - 3z_2 - 4z_3|$ is not less than (where $z_1, z_2, z_3$ are complex numbers)-
(A) 43 (B) 44 (C) 45 (D) 46

**10.** Consider complex numbers z and w satisfying the equation $z + w = \frac{\bar{z}}{w}$ and $\frac{1}{w} + z = w\bar{z}$, then which of the following is/are correct ?

(A) The value of $|z| + |w|$ is 2
(B) If $\arg(z) = \pi$ then, $\arg(w) = -\frac{\pi}{3}$ or $\frac{\pi}{3}$
(C) If $w = 1$, then $\text{Re}(z) = \frac{1}{2}$
(D) If $w = -1$, then $\text{Re}(z) = \frac{1}{2}$


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**11.** Given $z_1, z_2, z_3, z_4$ are four points in the complex plane such that $|z_1| < 1$, $|z_2| = 1$ and $|z_3| \le 1$ and $z_3 = \dfrac{z_2(z_1 - z_4)}{\bar{z}_1 z_4 - 1}$, then $|z_4|$ can be $(z_1 \ne z_4)$ -

(A) 2 (B) $\frac{2}{5}$ (C) $\frac{1}{3}$ (D) $\frac{5}{2}$

**Linked Comprehension Type (Single Correct Answer Type)** (3 Marks each, –1 for wrong answer)

**Paragraph for Question 12 to 14**

Let $z_1$ and $z_2$ be two complex numbers satisfying $(z - \omega)^2 + (z - \omega^2)^2 = 0$, where $\omega$ is non real cube root of unity. Also let '$\alpha$' be a variable point on the circle $\left|z + \frac{1}{2}\right| = \frac{\sqrt{3}}{2}$ and $\alpha_1$, $\alpha_2$ are values of $\alpha$ such that $|\alpha|$ is maximum and minimum respectively.

**12.** Value of $|z_1 - \alpha|^2 + |z_2 - \alpha|^2 + |\omega - \alpha|^2 + |\omega^2 - \alpha|^2$ is -

(A) 3 (B) $\sqrt{3}$ (C) 6 (D) 1

**13.** Which of following is incorrect ?

(A) $|\alpha_1| + |\alpha_2| = |\alpha_1 - \alpha_2|$ (B) $\alpha_1 = z_2$ and $\alpha_2 = z_1$ where $z_1 < 0, z_2 > 0$

(C) $|2z_1 + 1| = |2z_2 + 1|$ (D) $\text{amp}\left(\frac{z_1}{z_2}\right) = \text{amp}(z_1.z_2)$

**14.** Number of complex numbers $\alpha$ such that $|\alpha - \alpha_1| + |\alpha - \alpha_2| = \sqrt{6}$ is

(A) 4 (B) 3 (C) 2 (D) 0

**SECTION–III**

**Numerical Grid Type (Upto Second Decimal place)** (4 Marks each, –1 for wrong answer)

**1.** If $A = \left\{z : \left|\frac{z-2}{z+2}\right| = 3, z \in C\right\}$ and $z_1, z_2, z_3, z_4 \in A$ are 4 complex numbers representing points P, Q, R, S respectively on the complex plane such that $z_1 - z_2 = z_4 - z_3$, then maximum value of area of quadrilateral PQRS is

**2.** If $\alpha$ is a root of equation $x^2 - x + 1 = 0$, then the value of $|\alpha^{333} + \alpha^{666} + \alpha^{999}|$ is equal to -

**3.** A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 3 units and then vertically away from origin by 1 units to reach a point $z_1$. From $z_1$ the particle moves $\sqrt{2}$ units in the direction of the vector $-\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in clockwise direction on a circle with centre at origin, to reach a point $p + iq$. The value of $(2p + q)$ is

**4.** Consider the curves $C_1 : |z - 2| = 2 + \text{Re}(z)$ and $C_2 : |z| = 3$ (where $z = x + iy$, $x, y \in R$ and $i = \sqrt{-1}$). They intersect at P and Q in the first and fourth quadrants respectively. Tangents to $C_1$ at P and Q intersect the x-axis at R and tangents to $C_2$ at P and Q intersect the x-axis at S. If area of $\Delta PRS$ is $\lambda\sqrt{2}$ sq.units, then $(\lambda^2)$ is

**REVISION ASSINGMENT # 02 (VECTOR-3D-02) MATHEMATICS**

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|---|---|---|---|---|---|---|---|---|---|---|---|
| SECTION-I | Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| | A. | C | C | B | A | B | B | B | B | A | A |
| | Q. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| | A. | A,C | A,C,D | A,C | A,B,C,D | A,B,C | A,B,C | C | B | B | B |
| | Q. | 21 | 22 | 23 | 24 | | | | | | |
| | A. | C | D | A | A | | | | | | |
| SECTION-II | Q. | 1 | 2 | 3 | 4 | 5 | 6 | | | | |
| | A. | 1.25 | 1.00 | 2.00 | 10.80 | 0.85 or 0.86 | 4.50 | | | | |


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